

Optimal-Order so-Paving for Type-I MASAs

A scoped literature-implied answer to Conjecture 2.8(2)

Source: Popa–Vaes, arXiv:1412.0631v3. Support: Ravichandran–Srivastava, arXiv:1706.03737v2.

Status. `literature_implied_answer` (partial subcase). The implication below is not advertised as a theorem in the supporting paper. It results by replacing the matrix-paving input in Section 3 of the source paper by the later optimal multi-paving theorem.

Source question and location. Conjecture 2.8(2), page 8 of arXiv:1412.0631v3, asks whether there is a universal constant C such that

$$n_s(x, \varepsilon) \leq C\varepsilon^{-2}$$

for every self-adjoint x and every MASA $A \subset M$. Here n_s is the strong-operator paving size from Definition 2.2. Section 3 proves that every MASA in a type-I von Neumann algebra with separable predual is so-pavable, with the then-available estimate $O(\varepsilon^{-4})$.

Supporting theorem and location. Ravichandran–Srivastava, Theorem 1 on page 1 of arXiv:1706.03737v2, states that k zero-diagonal Hermitian contractions admit a common one-sided η -paving with at most $18k\eta^{-2}$ blocks. Applied to T and $-T$, this gives a two-sided paving

$$\max_j \|P_j T P_j\| < \eta \|T\|, \quad r \leq 36\eta^{-2}. \quad (1)$$

Literature-implied theorem. *Let M be a type-I von Neumann algebra with separable predual and let $A \subset M$ be an arbitrary MASA. For every $x = x^* \in M$ and $0 < \varepsilon < 1$,*

$$n_s(A \subset M; x, \varepsilon) \leq \lceil 144\varepsilon^{-2} \rceil. \quad (2)$$

Consequently Conjecture 2.8(2) has the optimal order ε^{-2} on the entire type-I class.

Proof. We follow exactly the type-I reduction and lifting argument of Section 3 in the source paper, changing only its finite-matrix estimate.

First consider a discrete diagonal field

$$\mathcal{D} \overline{\otimes} L^\infty(X) \subset \mathcal{B}(\ell^2 I) \overline{\otimes} L^\infty(X).$$

For finite I , (1) applies pointwise. For arbitrary countable I , apply the finite theorem to increasing finite corners and pass to a coloring in the compact product space $\{1, \dots, r\}^I$. For a measurable field, the successful colorings form a Borel subset of that compact space times X ; the measurable-selection step in Lemma 3.1 of the source paper therefore chooses them measurably, with the same r .

For arbitrary $x = x^*$ in such a field, put $a = E_A(x)$ and $T = x - a$. Then $E_A(T) = 0$, $\|a\| \leq \|x\|$, and $\|T\| \leq 2\|x\|$. Use (1) with $\eta = \varepsilon/2$. The resulting number of blocks is at most $36(\varepsilon/2)^{-2} = 144\varepsilon^{-2}$ and

$$\left\| \sum_j p_j (x - a) p_j \right\| = \max_j \|p_j T p_j\| \leq \frac{\varepsilon}{2} \|T\| \leq \varepsilon \|x\|. \quad (3)$$

Thus (2) even holds as a norm paving on every discrete expected component.

It remains to recall the continuous type-I component used in Proposition 3.2 of the source paper:

$$L^\infty([0, 1]) \overline{\otimes} L^\infty(X) \subset \mathcal{B}(L^2[0, 1]) \overline{\otimes} L^\infty(X). \quad (4)$$

Fix a strong neighborhood $\mathcal{V}$ of zero. Let q_m project onto the step functions that are constant on the m equal subintervals of $[0, 1]$; choose m so that $1 - q_m \otimes 1 \in \mathcal{V}$. If $V_m : \mathbb{C}^m \rightarrow L^2[0, 1]$ is the corresponding step-function isometry, compress x to

$$y = (V_m^* \otimes 1)x(V_m \otimes 1).$$

As in the source proof, let b be the diagonal of y and lift b to an element $a \in A$ that is constant on each interval. Then $\|a\| \leq \|x\|$, the matrix field $T = y - b$ is zero diagonal, and $\|T\| \leq 2\|x\|$.

Apply (1), again with $\eta = \varepsilon/2$, and choose the fiberwise partitions measurably. Lift their diagonal matrix projections to a partition $p_1, \dots, p_r \in A$. The computation in Proposition 3.2 gives

$$\left\| (q_m \otimes 1) \left(\sum_j p_j x p_j - a \right) (q_m \otimes 1) \right\| = \left\| \sum_j f_j(y - b) f_j \right\| \leq \varepsilon \|x\|. \quad (5)$$

The number $r \leq \lceil 144\varepsilon^{-2} \rceil$ is independent of $\mathcal{V}$, which is precisely the required so-paving estimate. The standard direct-sum decomposition of a type-I algebra and padding partitions with zero projections completes the proof. $\square$

Sharpness and scope. The order cannot be improved uniformly over type-I MASAs: the diagonal MASA in $\mathcal{B}(\ell^2\mathbb{N})$ already has the lower bound $\Omega(\varepsilon^{-2})$, as recalled in both papers. The theorem does not settle so-paving for a general non-type-I MASA and does not establish the universal constant requested by Conjecture 2.8 in that generality.

Search status. A bounded search on 2026-08-13 used the exact source title and the phrases “so-paving MASA”, “Popa Vaes so-paving conjecture”, and “strong operator paving MASA”. It found no claimed solution of the arbitrary-MASA conjecture. The supporting authors cite Popa–Vaes while discussing optimal multi-paving, but do not explicitly state (2); the relation is an agent-identified implication.

References.

- S. Popa and S. Vaes, *Paving over arbitrary MASAs in von Neumann algebras*, arXiv:1412.0631v3 (2015).
- M. Ravichandran and N. Srivastava, *Asymptotically Optimal Multi-Paving*, arXiv:1706.03737v2 (2017).
- P. Casazza, D. Edidin, D. Kalra, and V. Paulsen, *Projections and the Kadison–Singer problem*, Oper. Matrices 1 (2007), 391–408.